

INDUSTRIAL ACCIDENTS AND INJURIES

→ An industrial accident is an unexpected and unintentional event that disrupts work procedures and has the potential to cause damage to property and injury or death to workers.

→ A hazard is a condition or set of conditions that has the potential for causing an accident or other harmful outcome.

→ The danger posed by a hazard is the relative exposure or liability to injury, death, and/or damage from that hazard.

Three factors are commonly identified in the ergonomics literature as the primary reasons:

- **Human errors.** These factors relate to the worker or workers who are responsible for the operation and make mistakes that are sometimes the direct cause of an industrial accident.
- **Job factors.** This category refers to the kinds of tasks, methods, materials, equipment, and so on that are used in the operation. Some jobs are more dangerous than others.
- **Environmental conditions.** The environment includes lighting, noise, temperature, and other conditions that surround the operation. For example, the dimlighting in a plant made it difficult for a worker to see the obstacle in his path, and he tripped and broke his ankle.



Figure 1 A model depicting the possible factors that contribute to industrial accidents.

→ Why do industrial accidents occur? Three factors are commonly identified in the ergonomics literature as the primary reasons:

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1. Human Error

Human error is a major cause of industrial accidents, either directly or indirectly. It refers to improper or inadvertent human actions that reduce workplace safety.

- **Types:**
 - **Errors of Omission:** Failure to take required action (e.g., forgetting to turn off a circuit breaker).
 - **Errors of Commission:** Taking incorrect action (e.g., switching off the wrong circuit breaker).
- **Factors Affecting Accident Rates:**
 - **Age:** Young workers have higher accident rates due to risk-taking, while elderly workers struggle with physical tasks.
 - **Time on Job:** Most accidents occur in the first three years, peaking at 2–3 months.
 - **Fatigue & Stress:** Reduce awareness and focus, increasing accident risks.

2. Job Factors

Accident rates vary by industry due to inherent job risks.

- **Methods:**
 - Manual work increases fatigue and accidents.
 - Mechanized and automated systems reduce but do not eliminate risks (e.g., forklift errors, conveyor pinch points).
- **Equipment Hazards:**
 - **Electrical Hazards:** Shocks, burns, arc flashes.
 - **Mechanical Hazards:** Rotating machinery, shearing, pressing, impact forces causing injuries like cuts, fractures, and crushed limbs.
 - **Temperature Hazards:** Hot (welding, casting) or cold (cryogenic) equipment causing burns and skin damage.

3. Material Hazards

- **Corrosive Materials:** Acids, bases causing tissue damage.
- **Toxic/Irritant Materials:** Poisons affecting organs and functions.
- **Flammable Materials:** Require oxidizer, heat, and chemical reaction for ignition. Prevention involves separation, quenching, or inhibiting reactions.

4. Environmental Conditions

- **Physical Factors:** Poor lighting, noise, temperature, ventilation.
- **Social/Psychological Factors:** Work stress, peer pressure, unsafe work culture.

5. Accident Prevention Techniques

- **Machine Design:** Inaccessible dangerous parts.
- **Enclosures & Guards:** Isolating machines and hazardous operations.
- **Safety Features:** Interlocks, emergency stop buttons, two-hand start buttons.

OCCUPATIONAL DISORDERS AND DISEASES

→ The injury or health problem is one that results from a cumulative condition. These types of disorders, which may take years before their effects are evident, are called cumulative trauma disorders.

→ there are several causal factors leading to cumulative trauma disorders: (1) highly repetitive activity, (2) unnatural joint postures, and (3) application of forces through hinge joints.

→ Occupational diseases usually result from exposure to a substance over an extended period of time. In some cases, the onset of the disease is delayed for many years.